

Alex Zhang

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EDUCATION

Monash University <i>Bachelor of Computer Science (Distinction); Data Science Specialization</i>	Melbourne, Australia Feb. 2023 – Present
Waseda University <i>Bachelor of Engineering, Computer Science & Communications Engineering (Exchange)</i>	Tokyo, Japan Apr. 2026 – Aug. 2026
University of Texas at Austin <i>Bachelor of Science, Mathematics (Exchange)</i>	Austin, USA Aug. 2024 – Dec. 2024

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, Typescript, SQL
Frameworks & Libraries: PyTorch, NumPy, Pandas, React, Tailwind CSS
Infrastructure & Tools: AWS, Selenium, Git, Linux

EXPERIENCE

AI Research Intern <i>CSIRO</i>	Feb. 2026 – Present Australia
- Architecting an asynchronous, high-throughput Python ingestion engine to concurrently scrape and sanitize privacy policies and metadata across app marketplaces, eliminating a manual assessment bottleneck that historically required over 160 minutes per application. - Designing a distributed, multi-agent LLM orchestration layer to evaluate applications against the 50 distinct cybersecurity criteria of the ACSIMA framework, isolating metadata validation from legal text analysis to enforce strict context boundaries and prevent model hallucinations	
AI Intern <i>Hawthorn FC</i>	Aug. 2025 – Dec. 2025 Melbourne, Australia
- Engineered AI agents to process GPS and match-event telemetry, reducing data processing latency by 90% and eliminating manual workflows. - Streamlined the opposition analysis process by developing scripts to scrape and aggregate public league data, providing coaches with instant access to competitor metrics.	

PROJECTS

NASDAQ Feed Handler <i>C++, mmap, Low-Latency</i>
- Engineered a low-latency market data feed handler processing 11.5 million msg/s on a single core. - Implemented a Zero-Copy architecture using Memory Mapped Files (mmap) and reinterpret_cast to eliminate user-space buffer copies. - Optimized Order Book data structures by replacing Red-Black Trees (std::map) with contiguous memory vectors (std::vector) for cache locality, improving throughput by 65%.
Vector Vault <i>Python, NumPy, HNSW, K-Means</i>
- Engineered a custom vector search engine implementing the HNSW algorithm, utilizing multi-layered graph traversal to optimize nearest-neighbor convergence. - Implemented vectorized linear algebra operations using NumPy broadcasting and matrix operations to maximize CPU cache locality. - Developed a K-Means clustering algorithm from first principles to power a Product Quantization (PQ) pipeline, compressing 1024-dimensional float vectors into 8-bit codes.

LEADERSHIP & AWARDS

Accenture Tech Bootcamp <i>First Place Winner</i>	Feb. 2026
WIRED Monash <i>Technical Events Lead</i>	Oct. 2024 – May. 2025
Australia-China Youth Association (ACYA) <i>Case Competition Finalist</i>	Mar. 2024